

Toda función homogénea es subaditiva si y solo si es convexa

Ejercicio 1. Sea $F: \mathbb{R} \rightarrow \mathbb{R}$ homogénea, prueba que F es subaditiva $\iff F$ es convexa.

Solución: Veamos ambas implicaciones:

$\Rightarrow$ Sea $t \in [0, 1]$ y sean $x, y \in \mathbb{R}$. Por la subaditividad de F , tenemos que

$$F(tx + (1-t)y) \leq F(tx) + F((1-t)y) \stackrel{\text{homogénea}}{\leq} tF(x) + (1-t)F(y).$$

$\Leftarrow$ Sean $x, y \in \mathbb{R}$. Por la convexidad de F , tenemos que

$$F(x+y) = F\left(2 \cdot \frac{x+y}{2}\right) \stackrel{\text{homogénea}}{\leq} 2F\left(\frac{x+y}{2}\right) \leq 2 \cdot \left(\frac{1}{2}F(x) + \frac{1}{2}F(y)\right) = F(x) + F(y). \quad \blacksquare$$

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